

Unit-3

Data Science [ChatGpt]

2 Marks questions

Here are the detailed answers for your questions on **Data Warehousing** and **OLAP**, with explanations suitable for **4 or 6 mark questions**.

1) Define and Explain Data Warehouse

Definition:

A **Data Warehouse** is a **centralized repository** used for storing large volumes of data collected from multiple heterogeneous sources, structured for **querying and analysis** rather than transaction processing.

Explanation:

- It supports **decision-making** by providing **historical, integrated, subject-oriented, and non-volatile data**.
- Data is extracted, transformed, and loaded (ETL) from operational databases.

Example:

An e-commerce company uses a data warehouse to analyze customer purchases, sales trends, and inventory movement.

Key Characteristics (for 6 marks):

- **Subject-Oriented:** Organized around key subjects like customer, sales.
 - **Integrated:** Combines data from multiple sources.
 - **Time-Variant:** Contains historical data.
 - **Non-Volatile:** Data is stable and not changed frequently.
-

2) Explain Four/Six Differences Between Operational Database System and Data Warehouse

Operational DBMS	Data Warehouse
Used for daily operations	Used for analysis and decision-making
Supports INSERT, UPDATE, DELETE	Mainly supports SELECT (read-heavy)
Real-time, current data	Historical data
Normalized schema	Denormalized (star/snowflake) schema
Optimized for transactions	Optimized for read and analytical queries
Frequent data changes	Periodic data loading

Example:

- **Operational DB:** Banking system for deposits and withdrawals.
- **Data Warehouse:** Tracks financial performance over years.

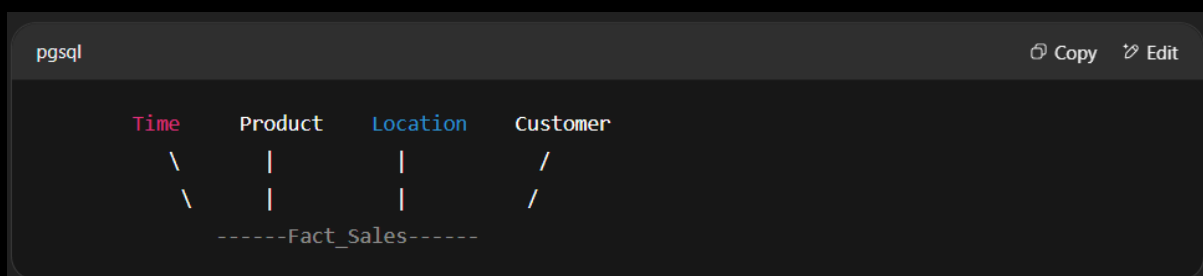
3) Explain the Following Multidimensional Database Schemas with Example

i) Star Schema

Definition:

A **Star Schema** is a type of multidimensional schema with a **central fact table** connected to multiple **dimension tables** in a star-like structure.

Example:



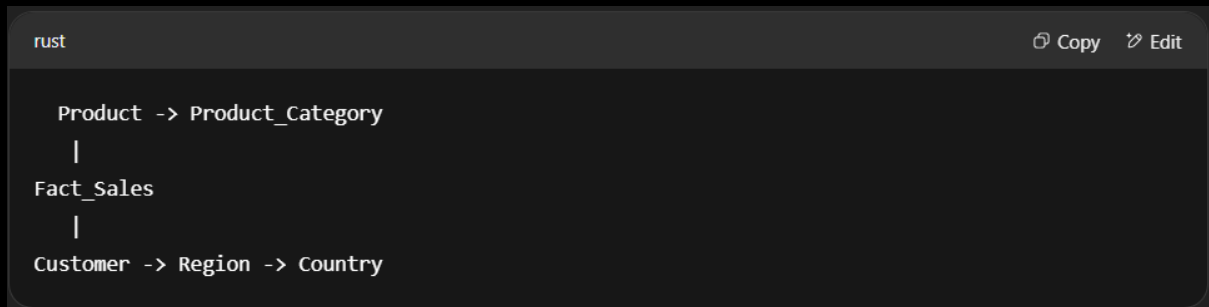
Use: Simple and fast query performance. Used in retail, banking, etc.

ii) Snowflake Schema

Definition:

An extension of star schema where **dimension tables are normalized**, leading to **multi-level relationships** (hierarchies).

Example:



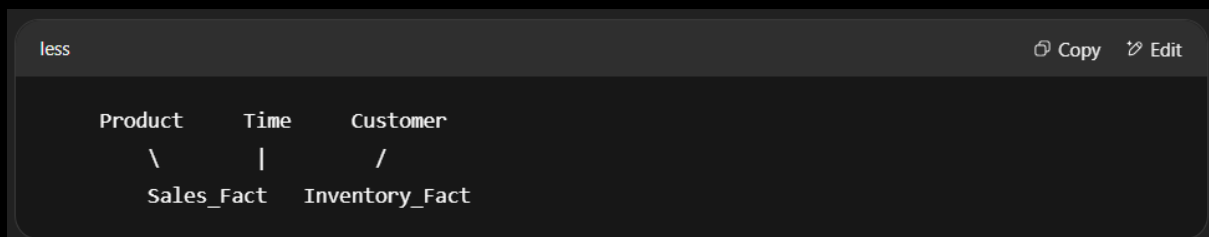
Use: Saves space, but complex joins may reduce performance.

iii) Fact Constellation Schema (Galaxy Schema)

Definition:

A schema with **multiple fact tables** sharing **dimension tables**. It represents a complex data warehouse design.

Example:



Use: Useful for complex business models involving multiple processes.

4) Define Measure. Explain Different Categories of Measures

Definition:

A **measure** is a **numeric value** in a fact table that can be analyzed and aggregated (e.g., sales amount, profit).

Types of Measures:

1. **Distributive:** Can be computed by partitioning (e.g., SUM, COUNT).
2. **Algebraic:** Can be derived using distributive measures (e.g., AVERAGE = SUM / COUNT).
3. **Holistic:** Cannot be broken into parts (e.g., MEDIAN, MODE).

Example:

- Sales revenue → distributive
- Average salary → algebraic

- Median order value → holistic
-

5) Explain Any Four/Six OLAP Operations

1. **Roll-Up:**
Aggregates data from lower to higher levels (e.g., Day → Month → Year).
2. **Drill-Down:**
Opposite of roll-up, moves from summary to detail (e.g., Year → Quarter → Month).
3. **Slice:**
Selects one dimension and fixes it at a value (e.g., sales in 2024 only).
4. **Dice:**
Selects a sub-cube by choosing multiple dimensions (e.g., products in Jan–Mar in region A).
5. **Pivot (Rotate):**
Reorients the cube (e.g., swapping rows and columns in a report).
6. **Drill-Across:**
Uses more than one fact table that shares dimensions.

Example:

Analyzing quarterly sales by region using roll-up, or filtering specific product categories using slice.

Here are the answers to **questions 6 to 10** from the **Data Warehouse** unit, written in a format suitable for **4 or 6 mark questions**, with explanation and examples where applicable.

6) Explain Different Views Regarding the Business Analysis Framework Design of a Data Warehouse

There are **four main views** used in business analysis to design a data warehouse:

1. **Top-Down View:**
 - Focuses on how data is used for decision-making.
 - Identifies key performance indicators (KPIs), reports, and metrics needed by management.
2. **Data Source View:**
 - Identifies operational databases, flat files, and other external data sources.
 - Helps understand where and how to collect data.

3. Data Warehouse View:

- Describes how data is stored in the warehouse (schemas like star/snowflake).
- Shows fact and dimension tables.

4. Business Query View:

- Defines how end users will interact with the data (e.g., OLAP tools, dashboards).
- Ensures data meets user query requirements.

7) Explain the Steps in Data Warehouse Design Process

Here are the key steps in designing a data warehouse:

1. Business Requirements Analysis:

- Understand what insights or reports the organization needs.
- Identify KPIs and end-user requirements.

2. Data Modeling:

- Choose schema (Star, Snowflake, or Fact Constellation).
- Define fact tables, dimension tables, and relationships.

3. ETL Design (Extract, Transform, Load):

- Design the process for extracting data from sources, cleaning it, transforming it, and loading into the warehouse.

4. Physical Design:

- Decide on storage structures, indexing, partitioning, and backup strategies.

5. Implementation & Testing:

- Build the warehouse, test data accuracy, and validate performance.

6. Deployment and Maintenance:

- Deploy to production and regularly update and maintain data quality.

8) Explain the Three-Tier Architecture of Data Warehouse with Neat Diagram

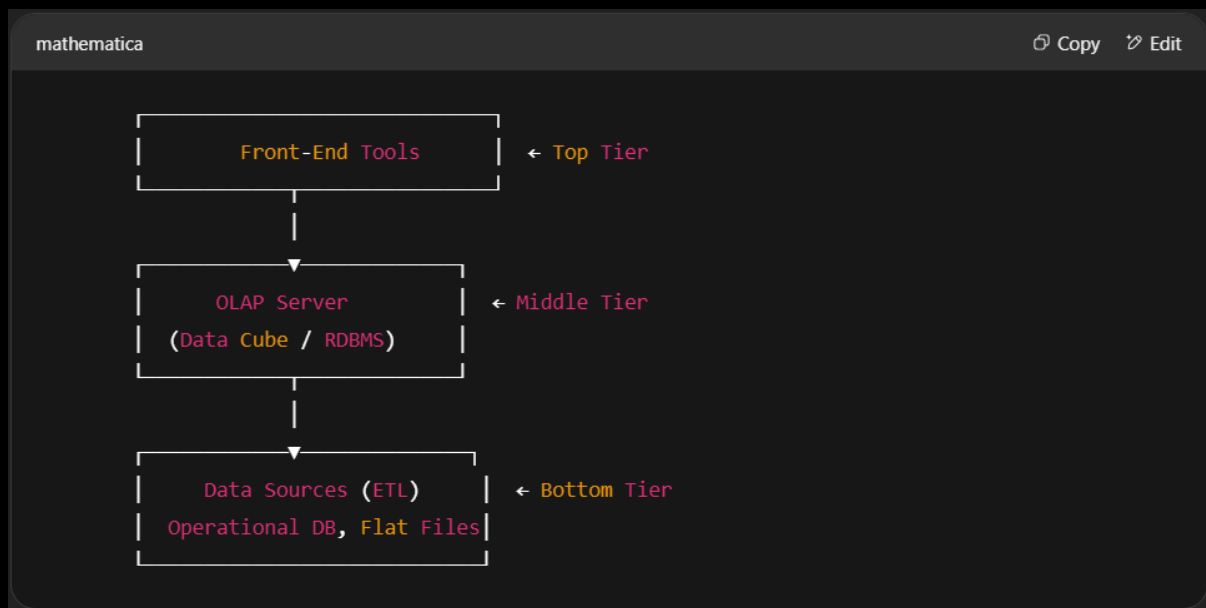
Three-Tier Architecture:

1. Bottom Tier – Data Source Layer:

- Contains operational databases, external sources, flat files.

- ETL processes extract and cleanse data from this tier.
2. **Middle Tier – Data Storage/OLAP Server:**
 - Houses the **data warehouse database** (relational or multidimensional).
 - OLAP server performs multidimensional analysis.
 3. **Top Tier – Front-End Tools/User Interface:**
 - Contains **reporting tools**, dashboards, query tools used by business users.

Diagram:



9) Explain the Different Data Warehouse Models from the Architecture Point of View

1. **Enterprise Warehouse:**
 - Centralized warehouse for the entire organization.
 - Stores historical data from all business processes.
2. **Data Mart:**
 - Subset of a data warehouse.
 - Focused on a specific department or business unit (e.g., marketing, sales).
3. **Virtual Warehouse:**
 - Provides a logical view without physically storing all the data.
 - Uses real-time integration and views across sources.

Example:

- A marketing data mart stores only customer and campaign data.

- A virtual warehouse may pull live data from CRM and sales databases.
-

10) Explain the Different Back-End Tools and Utilities Included in Data Warehouse

Key back-end tools and utilities:

1. Data Extraction Tools:

- Pull data from various sources (databases, flat files, APIs).

2. Data Cleaning Tools:

- Detect and correct errors, inconsistencies, or missing values in data.
- Example: Data scrubbing tools.

3. Data Transformation Tools:

- Convert data into suitable formats, apply rules, perform normalization.

4. Load Utilities:

- Load the processed data into the data warehouse tables.
- Support batch and real-time loading.

5. Metadata Repository Tools:

- Store and manage metadata (data about data), such as table names, data types, and transformation rules.

Explanation:

These tools ensure **data quality**, **efficient ETL**, and **easy management** of data pipelines in the warehouse environment.

Here are the answers for **questions 11 to 17** related to **Data Warehousing and Preprocessing**, written in a format suitable for **4 or 6 mark questions** with definitions, explanations, and examples.

11) What is Metadata? Explain Metadata Repository

Metadata:

- Metadata is "data about data".
- It describes the structure, operations, and properties of the data in the data warehouse.

Types of Metadata:

1. **Technical Metadata:** Describes data models, schemas, source structures, ETL processes.
2. **Business Metadata:** Describes business meaning of data (e.g., "Customer_ID" means unique identifier for a customer).
3. **Operational Metadata:** Includes data loading schedules, error logs, data lineage.

Metadata Repository:

- A **centralized location** where metadata is stored and managed.
- Helps users understand how data was derived, formatted, and used.
- Provides details like:
 - Data source name
 - Field type and length
 - Transformation rules
 - Load frequency

Example:

When a user accesses a sales report, the repository shows that “Total Sales” is calculated from invoice data transformed during ETL.

12) Explain Different Types of OLAP Servers

OLAP Servers:

1. **Relational OLAP (ROLAP):**
 - Works on relational databases.
 - Uses SQL queries to perform OLAP operations.
 - Handles large volumes of data but is slower than MOLAP.
2. **Multidimensional OLAP (MOLAP):**
 - Uses pre-computed data cubes.
 - Fast query performance.
 - Limited scalability.
3. **Hybrid OLAP (HOLAP):**
 - Combines ROLAP and MOLAP.
 - Frequently used data is stored in cubes (MOLAP), and detailed data is stored in relational tables (ROLAP).

4. Desktop OLAP (DOLAP):

- OLAP engine runs on the user's desktop.
- Uses local data cubes for small-scale data analysis.

Diagram (optional for 6 marks) can show architecture layers with users accessing OLAP servers.

13) Explain the Different Forms of Data Preprocessing

Data preprocessing prepares raw data for analysis. Main forms include:

1. Data Cleaning:

- Handles missing values, noisy data, and inconsistencies.
- E.g., fixing typos or removing duplicates.

2. Data Integration:

- Combines data from multiple sources.
- E.g., merging customer data from CRM and sales databases.

3. Data Transformation:

- Converts data into a suitable format for analysis.
- Includes normalization, aggregation, etc.

4. Data Reduction:

- Reduces volume but keeps the same analytical quality.
 - E.g., dimensionality reduction, data cube aggregation.
-

14) Explain Any Four Ways of Handling Missing Values

1. Ignore the Record:

- Drop records with missing values.
- Useful when dataset is large and missing data is small.

2. Fill with Global Constant:

- Replace missing value with a fixed value like "Unknown" or 0.

3. Fill with Mean/Median/Mode:

- Replace missing values with the average or most frequent value of the attribute.

4. Predict Using Other Attributes:

- Use regression or classification models to predict the missing value.

Example:

If a student's "Grade" is missing, it could be filled with the average of the class or predicted based on attendance and test scores.

15) Define Noise and Explain Different Data Smoothing Techniques

Noise:

- Noise is **random error or irrelevant information** in data that can distort analysis.

Data Smoothing Techniques:

1. Binning:

- Sort values and divide into bins.
- Replace each bin with mean, median, or boundary values.

2. Regression:

- Use regression functions (linear or non-linear) to fit data and smooth out variations.

3. Clustering:

- Group similar data and replace with cluster representative (mean or mode).

4. Moving Average:

- Replace each data point with the average of neighboring values to reduce fluctuations.

Example:

In sales data, a sudden spike can be smoothed using moving averages over a week.

16) Explain the Different Steps Involved in Data Transformation

1. Smoothing:

- Removes noise using methods like binning or regression.

2. Aggregation:

- Summarizes data (e.g., daily to monthly totals).

3. Generalization:

- Replaces detailed data with higher-level concepts (e.g., ZIP code → city).

4. **Normalization:**

- Scales data into a standard range (e.g., 0 to 1).

5. **Attribute Construction:**

- Creates new attributes from existing ones (e.g., BMI from height and weight).
-

17) What is Data Reduction? Explain the Strategies for Data Reduction

Data Reduction:

- The process of reducing the **volume of data** while maintaining its **analytical integrity**.

Strategies:

1. **Dimensionality Reduction:**

- Removes irrelevant attributes using methods like PCA or feature selection.

2. **Numerosity Reduction:**

- Replaces data with a smaller form (e.g., regression models, histograms, clustering).

3. **Data Compression:**

- Encodes data efficiently using techniques like Huffman coding or wavelet transforms.

4. **Data Cube Aggregation:**

- Aggregates data across dimensions (e.g., daily sales → monthly sales).

Example:

A 1 TB dataset can be reduced to 100 GB using attribute subset selection and aggregation without losing insights.

18) Write a Note on Wavelet Transform and Principal Component Analysis (PCA)

Wavelet Transform:

- It is a signal processing technique used to transform data into different frequency components.
- Applied to compress, denoise, or simplify data by transforming it into the wavelet domain.

How it Works:

- Wavelet transform decomposes data into approximation (low frequency) and detail (high frequency) components.
- Only the approximation part may be stored for data reduction.

Example:

- Applied in image compression or reducing time-series data.
-

Principal Component Analysis (PCA):

- A statistical technique for **dimensionality reduction**.
- Transforms high-dimensional data into a smaller set of uncorrelated components (called principal components) that retain most of the variance.

Steps:

1. Standardize the dataset.
2. Compute the covariance matrix.
3. Find eigenvectors and eigenvalues.
4. Select top k principal components.

Example:

- Reducing 10 features in a customer dataset to 2 principal components for visualization.
-

19) Explain Any Two Numerosity Reduction Techniques

Numerosity reduction techniques reduce the size of data by replacing the data with a **smaller representation**.

1. **Parametric Methods (e.g., Regression):**
 - Use a mathematical model (like linear regression) to represent the data.
 - Only the parameters (coefficients) of the model are stored.
2. **Clustering:**
 - Data points are grouped into clusters.
 - Each cluster is represented by its centroid or prototype.

Example:

- In a 1 million-point dataset, 10,000 clusters could represent patterns more compactly.

20) Explain Any Three Methods for Data Discretization

Data Discretization converts continuous data into discrete intervals.

1. Equal Width Binning:

- Divides the range of values into **equal-sized intervals**.
- Simple but may leave some bins empty if data is skewed.

2. Equal Frequency Binning:

- Each bin has approximately the **same number of data points**.
- Better for handling skewed distributions.

3. Cluster Analysis:

- Groups similar data points using clustering algorithms (like k-means).
- Each cluster acts as a bin.

Example:

- Continuous age values (18-65) may be binned into “Young”, “Middle-aged”, “Senior”.

21) Note on Methods Used for Generation of Concept Hierarchies for Categorical Data

Concept hierarchies provide **generalization paths** for categorical attributes (e.g., City → State → Country).

Methods:

1. Schema-Based:

- Defined in database schema using foreign key relationships.
- E.g., City → District → State → Country.

2. User-Defined:

- Predefined by users or domain experts based on domain knowledge.
- E.g., Course → Subject → Department.

3. Data-Driven (Dynamic):

- Generated by analyzing data patterns (e.g., clustering values into similar groups).

4. Specification in Ontologies or Taxonomies:

- Use knowledge bases like WordNet for creating concept trees.
-

22) Explain Data Cube Aggregation

Data Cube Aggregation is a **data reduction** technique that aggregates data across multiple dimensions.

Example:

- Instead of storing daily sales for every product in every city, aggregate it into **monthly sales by region**.

Benefits:

- Reduces data volume.
- Helps identify trends at higher conceptual levels.

OLAP Cubes:

- Commonly used in OLAP systems for fast multidimensional analysis (e.g., Sales by Region → Time → Product).
-

23) Explain Heuristic Methods of Attribute Subset Selection with Example

Attribute Subset Selection finds a subset of relevant features by removing irrelevant or redundant ones.

Heuristic Methods:

1. Stepwise Forward Selection:

- Start with no attributes.
- Add attributes one by one based on accuracy gain.

2. Stepwise Backward Elimination:

- Start with all attributes.
- Remove one attribute at a time that reduces accuracy the least.

3. Decision Tree Induction:

- Selects attributes that split data best (based on entropy or Gini index).

Example:

- From 10 attributes in a student dataset, only Attendance, Internal Marks, and Assignment Score may be selected for predicting performance.
-

24) What is Sampling? Explain the Ways Used to Sample for Data Reduction

Sampling:

- Sampling selects a **representative subset of data** from a large dataset.
- Used when working on full data is time-consuming or impractical.

Types of Sampling:

1. Simple Random Sampling:

- Every item has an equal chance of being selected.

2. Stratified Sampling:

- Divides data into strata (groups) and samples from each group proportionally.
- Ensures all categories are represented.

3. Systematic Sampling:

- Select every *k-th* item from a list.

4. Cluster Sampling:

- Randomly selects entire groups (clusters), not individual items.

Example:

- Sampling 1,000 transactions from a million records for model testing.
-